

NORTH SYDNEY BOYS HIGH SCHOOL

2016 HSC ASSESSMENT TASK 1

Mathematics

General Instructions

- Reading time – **5 minutes**
- Working time – **55 minutes**
- Write in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- Show all necessary working in every question
- Each new question is to be started on a **new page**.
- Use the **Reference Sheet** provided!

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Ireland
- ☐ Mr Jomaa
- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Ms Ziazaris
- ☐ Mr Zuber

Student Number: _____

(To be used by the exam markers only.)

Question No	1-5	6	7	8	9	Total	Total %
Mark	$\frac{5}{5}$	$\frac{17}{17}$	$\frac{16}{16}$	$\frac{4}{4}$	$\frac{14}{14}$	$\frac{56}{56}$	$\frac{100}{100}$

Section I: Objective Response (5 marks)

Mark your answers on the multiple choice box below.

Marks

- 1 Evaluate $\sum_{n=2}^5 (-1)^n n^2$ 1
 (A) 54 (B) 14 (C) -54 (D) -14

- 2 For what values of k does the equation $x^2 - 6x - 3k = 0$ have real roots? 1
 (A) $k \geq -3$ (B) $k \leq -3$ (C) $k \geq 3$ (D) $k \leq 3$

- 3 If the 5th and 18th terms of an arithmetic sequence are 12 and 64 respectively, then the common difference of the sequence is: 1
 (A) -5 (B) 4 (C) -4 (D) 5

- 4 A parabola has equation $x^2 = 12(8 - y)$. Its directrix is: 1
 (A) $y = -3$ (B) $y = 3$ (C) $y = 5$ (D) $y = 11$

- 5 The roots of the quadratic equation $gx^2 - x + h = 0$ are -1 and 3. 1
 The value of h is:
 (A) -6 (B) -3 (C) $-\frac{3}{2}$ (D) 2

Select the alternative A, B, C or D that best answers the question above and indicate your choice by shading the appropriate letter.

1.	☐ A	☐ B	☐ C	☐ D
2.	☐ A	☐ B	☐ C	☐ D
3.	☐ A	☐ B	☐ C	☐ D
4.	☐ A	☐ B	☐ C	☐ D
5.	☐ A	☐ B	☐ C	☐ D

Section I: Extended Responses

Write your solutions in your exam booklet, starting a new page for each Question.

Question 6 – Quadratics (17 marks)**Marks**

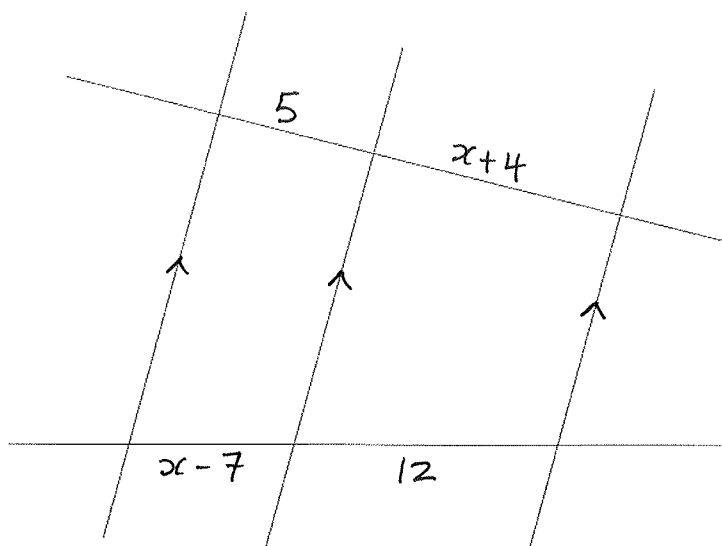
- (a) Find values of a, b, c such that $3x^2 - 7x - 6 \equiv a(x-2)^2 + b(x-2) + c$ 2
- (b) Solve the inequality $2x^2 + x - 6 \geq 0$. 2
- (c) The quadratic equation $x^2 - 7x + 8 = 0$ has roots α and β .
Find:
- (i) $\alpha + \beta$ 1
 - (ii) $\alpha\beta$ 1
 - (iii) $\frac{2}{\alpha} + \frac{2}{\beta}$ 1
 - (iv) $\alpha^2 + \beta^2$ 2
- (d) Solve the equation $m^4 + 12m^2 - 64 = 0$. 3
- (e) The line $y = x + k$ is tangent to the parabola $y = x^2 - 3x - 5$.
If P is the point where they meet,
- (i) Show that the x -coordinate of P satisfies $x^2 - 4x - 5 - k = 0$. 1
 - (ii) Find the value of k . 2
 - (iii) Find the coordinates of P . 2

Question 7 – Series (16 marks)**Marks**

- (a) The first 3 terms of an arithmetic progression are 7, 11, and 15.
- (i) Is 111 a term in this sequence? Justify your answer. 2
 - (ii) Find the sum of the first 26 terms. 2
- (b) Farmer Joe is trying to fill his 1200 litre water tank. 3
On day 1, he puts 100 litres into the tank.
On day 2 he adds 90 litres to the tank.
On day 3 he puts in 81 litres.
If he continues to add water in this way, will he ever fill the tank?
Justify your answer.
- (c) The sum of the first n terms of a sequence is given by $S_n = n(38 - 2n)$
- (i) Calculate S_1 and S_2 . 2
 - (ii) Find the first three terms of this series, T_1, T_2 , and T_3 . 2
 - (iii) Find an expression for the n^{th} term, T_n , giving your answer in simplified form. 2
- (d) The terms $x-1, x+1, x^2-1$ are in geometric progression. Find the three terms of the progression. 3

Question 8 – Geometry (4 marks)

Marks



Calculate the value of x , giving a geometrical reason.

4

Question 9 – Locus and Parabola (14 marks)

Marks

(a) Given a parabola with equation $x^2 - 6x - 7 = 8y$,

- | | |
|--|----------|
| (i) Find the coordinates of the vertex | 2 |
| (ii) Find the coordinates of the focus | 1 |
| (iii) Find the length of the latus rectum chord | 1 |
| (iv) Sketch the parabola, showing its focus, vertex and directrix. | 2 |

(b) A point $P(x, y)$ moves so that its distance from the point $A(6, 1)$ is always twice its distance from the point $B(-3, 4)$.

- | | |
|--|----------|
| (i) Show that the locus of P is a circle. | 2 |
| (ii) Write down the centre and radius of this circle | 1 |

(c) A point $P(x, y)$ moves so that it is equally distant from the point $A(-2, 0)$ and the line $x = -6$. Write down the equation of the locus of P . **2**

(d) A sequence of parabolas is defined as follows:

$$y = (ax - 1)^2, y = (2ax - 1)^2, y = (3ax - 1)^2, y = (4ax - 1)^2, \dots$$

- | | |
|---|----------|
| (i) What is the focal length of the 20 th parabola in this sequence? | 2 |
| (ii) What point do the vertices of these parabolas approach? | 1 |

END OF EXAM

Section I - Objective response

$$\begin{aligned} \boxed{1} \quad \sum_{n=2}^5 (-1)^n n^2 &= (-1)^2 \cdot 2^2 + (-1)^3 \cdot 3^2 + (-1)^4 \cdot 4^2 + (-1)^5 \cdot 5^2 \\ &= 4 - 9 + 16 - 25 \\ &= -14 \end{aligned} \quad \therefore D$$

$$\begin{aligned} \boxed{2} \quad \text{Real roots} &\Rightarrow \Delta \geq 0 \\ \therefore 36 - 4(1)(-3k) &\geq 0 \\ 36 + 12k &\geq 0 \\ k &\geq -3 \end{aligned} \quad \therefore A$$

$$\begin{aligned} \boxed{3} \quad a + 4d &= 12 \\ a + 17d &= 64 \\ \therefore 13d &= 52 \quad \therefore d = 4 \end{aligned} \quad \therefore B$$

$$\begin{aligned} \boxed{4} \quad x^2 &= 12(8-y) \\ \therefore x^2 &= -12(y-8) \\ x^2 &= -4 \cdot 3 \cdot (y-8) \\ \therefore V &= (0, 8), a=3, \text{concave down} \\ \therefore \text{directrix is } y &= 11 \\ \text{---} \text{---} \text{---} y &= 11 \end{aligned} \quad \therefore D$$


$$\begin{aligned} \boxed{5} \quad \text{Sum of roots} &\Rightarrow -1 + 3 = \frac{-(-1)}{g} \\ 2 &= \frac{1}{g} \quad \therefore g = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{Product of roots} &\Rightarrow (-1)(3) = \frac{h}{g} \\ -3 &= \frac{h}{g} \\ \therefore h &= -3g = -\frac{3}{2} \end{aligned} \quad \therefore C$$

6

$$(a) \quad 3x^2 - 7x - 6 \equiv a(x-2)^2 + b(x-2) + c$$

$$\text{Sub } x=2 \Rightarrow 3 \cdot 4 - 7 \cdot 2 - 6 = 0 + 0 + c$$

$$-8 = c$$

$$\text{Coeff. of } x^2 \text{ term} \Rightarrow a = 3$$

$$\text{Sub. } x=0 \Rightarrow 0 - 0 - 6 = 4a - 2b + c$$

$$-6 = 12 - 2b - 8$$

$$-10 = -2b \quad \therefore b = 5$$

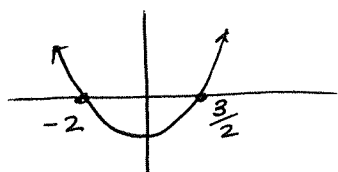
$$\text{Thus } a=3, \quad b=5, \quad c=-8.$$

✓ Significant progress

✓ Correct solution

$$(b) \quad 2x^2 + x - 6 \geq 0$$

$$(2x-3)(x+2) \geq 0$$



$$\therefore x \leq -2$$

$$\text{or } x \geq \frac{3}{2}$$

✓✓

$$(c) \quad x^2 - 7x + 8 = 0$$

$$(i) \quad \alpha + \beta = 7$$

$$(ii) \quad \alpha\beta = 8$$

$$(iii) \quad \frac{2}{\alpha} + \frac{2}{\beta} = \frac{2\alpha + 2\beta}{\alpha\beta} = \frac{2(\alpha + \beta)}{\alpha\beta} = \frac{14}{8}$$

$$= \frac{7}{4}$$

$$(iv) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= 7^2 - 2(8)$$

$$= 49 - 16$$

$$= 33$$

✓

✓

✓

✓

✓

6-continued

(d) $m^4 + 12m^2 - 64 = 0$

$\therefore (m^2 + 16)(m^2 - 4) = 0$

$\therefore m^2 + 16 = 0 \rightarrow \text{no solution}$

or $m^2 - 4 = 0 \rightarrow m = \pm 2.$

(e) (i) $y = x + k$ meets $y = x^2 - 3x - 5$

when $x + k = x^2 - 3x - 5$

$\therefore x^2 - 3x - 5 - x - k = 0$

ie. $x^2 - 4x - 5 - k = 0 \quad (*)$

(ii) Tangent $\Rightarrow \Delta = 0$

$\therefore 4^2 - 4 \cdot 1 \cdot (-5 - k) = 0$

$4 - (-5 - k) = 0$

$9 + k = 0$

$\therefore k = -9$

(iii) Equation (*) becomes

$x^2 - 4x - 5 + 9 = 0$

$x^2 - 4x + 4 = 0$

$(x - 2)^2 = 0$

$\therefore x = 2,$

$y = 2 - 9 = -7$

$\therefore P = (2, -7)$

7

(a) $T_1 = 7, T_2 = 11, T_3 = 15$

(i) $\therefore a = 7$
 $d = 4$

Then $111 = 7 + (n-1)4$

$\therefore n-1 = \frac{104}{4} = 26, n = 27.$

Integer, $\therefore 111$ is a term.

(ii) $S_{26} = \frac{26}{2} (a + l)$ where $l = 111 - 4$
 $= 107$

$\therefore S_{26} = \frac{26}{2} (7 + 107)$
 $= 1482$

(b) Amount added to tank is

$S = 100 + 90 + 81 + \dots$

This is a G.P. with $a = 100, r = 0.9$

$S_{\infty} = \frac{a}{1-r} = \frac{100}{1-0.9} = 1000$

Since the limiting sum 1000 L is less than the tank's capacity, it will not be filled.

7-continued

(c) $S_n = n(38 - 2n)$

(i) $S_1 = 1(38 - 2) = 36$

$S_2 = 2(38 - 4) = 68$

so $S_1 = 36$
 $S_2 = 68$

(ii) $T_1 = S_1 = 36$

$T_2 = S_2 - S_1 = 68 - 36 = 32$

Since $T_2 - T_1 = 32 - 36 = -4$

$\therefore T_3 = 32 - 4 = 28$

so $T_1 = 36$
 $T_2 = 32$
 $T_3 = 28$

(iii) $T_n = S_n - S_{n-1}$

$= n(38 - 2n) - (n-1)[38 - 2(n-1)]$

$= 38n - 2n^2 - [(n-1)(40 - 2n)]$

$= 38n - 2n^2 - [40n - 2n^2 - 40 + 2n]$

$= 38n - 2n^2 - 40n + 2n^2 + 40 - 2n$

$\therefore T_n = 40 - 4n$

(d) $x-1, x+1, x^2-1$ are in G.P.

$\therefore \frac{x+1}{x-1} = \frac{x^2-1}{x+1}$

$\therefore \frac{x+1}{x-1} = \frac{(x+1)(x-1)}{(x+1)}$

$\therefore x+1 = x^2 - 2x + 1$

$x^2 - 3x = 0$

$\therefore x = 0 \text{ or } x = 3$

So the 3 terms are $-1, 1, -1$

OR $2, 4, 8.$

✓✓

✓✓

✓ uses relation

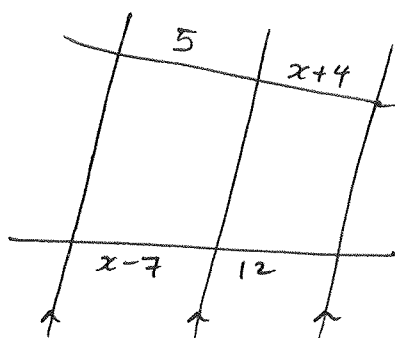
✓

✓ uses definition

✓

✓

8



$$\frac{x+4}{5} = \frac{12}{x-7} \quad \left(\text{Ratio of intercepts made by transversals on parallel lines} \right)$$

✓ equation
✓ reason

$$\therefore (x-7)(x+4) = 60$$

$$x^2 - 3x - 28 - 60 = 0$$

$$x^2 - 3x - 88 = 0$$

$$(x-11)(x+8) = 0$$

$$\therefore x = 11 \text{ or } -8$$

$$\text{But } x \text{ is a length, } \therefore x = 11.$$

✓✓
4

9

$$(a) (i) x^2 - 6x - 7 = 8y$$

$$\therefore x^2 - 6x + 9 - 9 - 7 = 8y$$

$$(x-3)^2 = 8y + 16$$

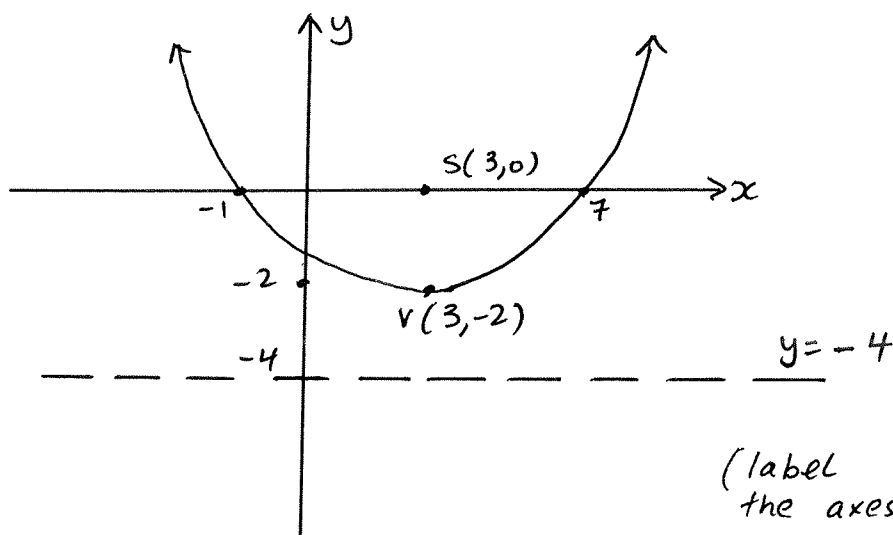
$$\therefore (x-3)^2 = 8(y+2)$$

$$\therefore \text{Vertex} = (3, -2)$$

$$(ii) a = 2 \therefore S = (3, 0)$$

$$(iii) \text{Length of latus rectum} = 4a = 8$$

(iv)



(label the axes!!)

✓✓

✓

✓

✓✓

[9]-continued

(b) $AP = 2 \times AB$

(i) $AP^2 = 4 \times AB^2$

$$\begin{cases} P = (x, y) \\ A = (6, 1) \\ B = (-3, 4) \end{cases}$$

$$\begin{aligned} \therefore (x-6)^2 + (y-1)^2 &= 4[(x+3)^2 + (y-4)^2] \\ &= 4[x^2 + 6x + 9 + y^2 - 8y + 16] \end{aligned}$$

$$x^2 - 12x + 36 + y^2 - 2y + 1 = 4x^2 + 24x + 36 + 4y^2 - 32y + 64$$

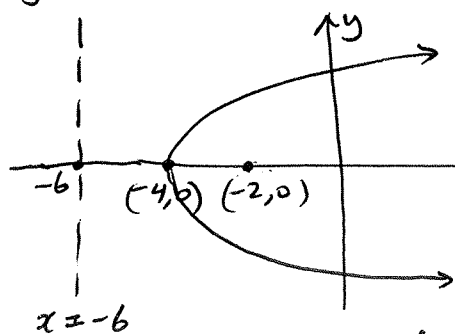
$$3x^2 + 36x + 3y^2 - 30y + 63 = 0$$

$$x^2 + 12x + 36 + y^2 - 10y + 25 = -21 + 36 + 25$$

$$(x+6)^2 + (y-5)^2 = 40$$

(ii) This is the equation of a circle with
centre $(-6, 5)$, radius $= \sqrt{40}$
 $= 2\sqrt{10}$

(c) By definition, this is a parabola with
focus $(-2, 0)$ and directrix $x = -6$:



i.e. focal length $a = 2$,
 $V = (-4, 0)$

$$\therefore y^2 = 8(x+4).$$

(d) (i) Following the pattern, T_{20} is $y = (20ax - 1)^2$
i.e. $y = [20a(x - \frac{1}{20a})]^2 = 400a^2(x - \frac{1}{20a})^2$

$$\text{Rearrange} \rightarrow (x - \frac{1}{20a})^2 = 4 \cdot \frac{1}{1600a^2} \cdot y$$

$$\text{So the focal length} = \frac{1}{1600a^2}$$

(ii) The vertices approach $(0, 0)$.